

Problem

Let

$$0 < a_1 < a_2 < \cdots < a_{mn+1}$$

be $mn + 1$ integers. Prove that one can select either

- $m + 1$ of them such that no one divides another, or
- $n + 1$ of them such that each divides the next.

Source: 1966 Putnam Mathematical Competition

§1 Revealing the Hidden Poset Structure

The divisibility relation naturally defines a partially ordered set on the given integers. The problem asks us to show that among $mn + 1$ elements there must exist either a long divisibility chain or a large collection of mutually incomparable elements.

To make this precise, for each index i define n_i to be the length of the longest divisibility chain

$$a_i = b_1 \mid b_2 \mid \cdots \mid b_{n_i},$$

where every b_j belongs to the set

$$\{a_i, a_{i+1}, \dots, a_{mn+1}\}.$$

Thus n_i records the maximum possible length of a chain beginning with a_i .

§2 The Critical Invariant

Suppose first that some n_i satisfies

$$n_i \geq n + 1.$$

Then there exists a chain

$$a_{i_1} \mid a_{i_2} \mid \cdots \mid a_{i_{n+1}},$$

consisting of $n + 1$ selected integers, each dividing the next. In this case the second conclusion of the problem already holds.

Therefore, to obtain a contradiction, assume that

$$n_i \leq n \quad \text{for all } i.$$

The values n_i can then take only the n possible values

$$1, 2, \dots, n.$$

Since there are $mn + 1$ integers and only n possible values of n_i , the pigeonhole principle implies that some value occurs at least

$$\left\lceil \frac{mn + 1}{n} \right\rceil = m + 1$$

times.

Hence there exist distinct indices

$$i_1 < i_2 < \cdots < i_{m+1}$$

such that

$$n_{i_1} = n_{i_2} = \cdots = n_{i_{m+1}}.$$

§3 Extracting the Hidden Antichain

We claim that the corresponding integers

$$a_{i_1}, a_{i_2}, \dots, a_{i_{m+1}}$$

are pairwise incomparable under divisibility.

Assume, for contradiction, that

$$a_{i_r} \mid a_{i_s}$$

for some $i_r < i_s$.

Since n_{i_s} is the length of the longest chain beginning at a_{i_s} , we may append a_{i_r} to the beginning of such a chain. This produces a divisibility chain beginning at a_{i_r} whose length is

$$n_{i_s} + 1.$$

Consequently,

$$n_{i_r} \geq n_{i_s} + 1,$$

which contradicts

$$n_{i_r} = n_{i_s}.$$

Therefore no one of the selected integers divides another.

We have thus found $m + 1$ integers forming an antichain.

§4 The Final Conclusion

Either some n_i is at least $n + 1$, yielding a divisibility chain of length $n + 1$, or every n_i is at most n , in which case the pigeonhole principle produces $m + 1$ integers no one of which divides another.

Hence among any $mn + 1$ positive integers

$$0 < a_1 < a_2 < \cdots < a_{mn+1},$$

one can always select either

$n + 1$ integers each dividing the next

or

$m + 1$ integers no one of which divides another.

This completes the proof.